Engine Model(s): PT6A

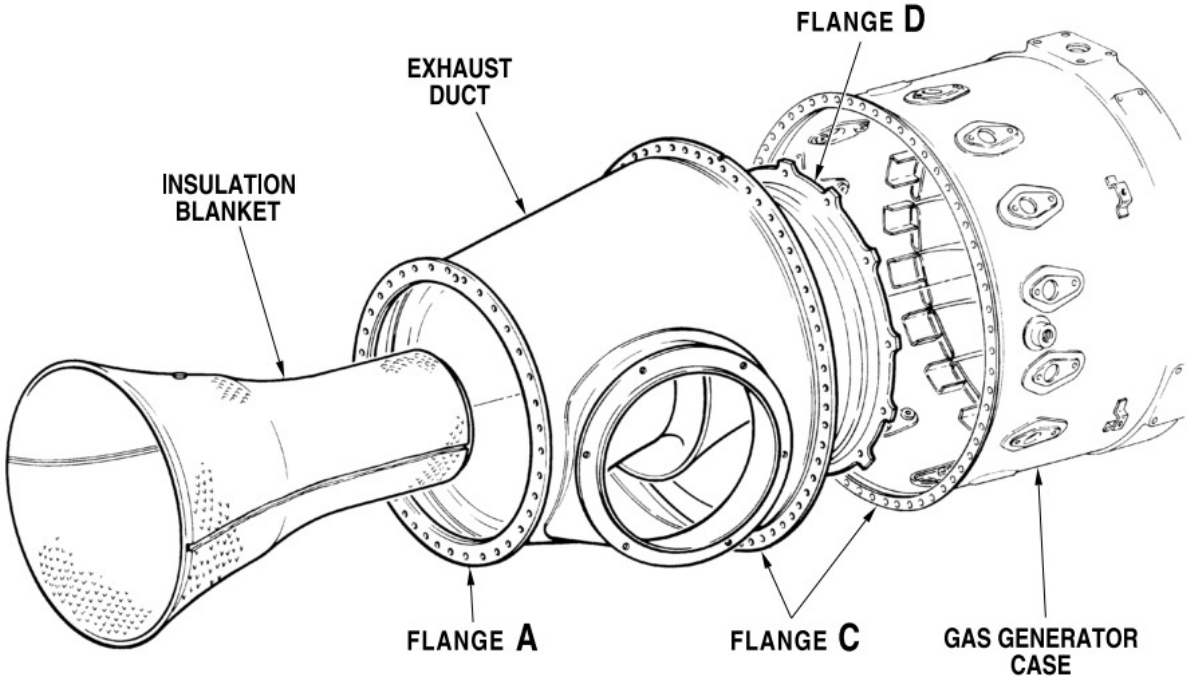
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EXHAUST DUCT - DESCRIPTION AND OPERATION

1. Description and Operation (Ref. Fig. 1)

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Figure 1 Exhaust Duct and Insulation Blanket



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The exhaust duct assembly consists of a heat-resistant nickel-alloy sheet, truncated cone fabrication with twin opposed exhaust ports and an annular sheet metal inlet. The exhaust port flanges provide the mounting for the airframe-supplied exhaust nozzles. Exhaust gases from the power turbine area enter the annular inlet around a coned inner sheet metal fabrication designed to impart gas swirl and to scavenge the duct with maximum efficiency.

The front flange of the duct (Flange A) provides the support and mounting flange for the reduction gearbox. Two flanges are provided at the rear of the duct; the inner flange (Flange D) provides the mounting for the power turbine stator housing, while the outer flange (Flange C) forms the major dividing flange between the gas generator assembly and the power section.

A dimpled, stainless steel/ceramic fiber laminate insulation blanket is installed between the inner cone of the duct and the power turbine shaft housing to minimize transfer of heat from the exhaust gases to the power turbine shaft bearings and rear section of the reduction gearbox.

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